

Aleena Roy

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EDUCATION:

Florida Atlantic University
Masters in Computer Science

Visvesvaraya Technological University
Bachelor of Engineering – Computer Science and Engineering

INTERNSHIPS:

Florida Atlantic University, Dania Beach | Graduate Research Assistant | Aug 2024 – December 2024

- Integrated YOLOv9 with ROS2 for real-time image capture and detection using the ZED2i camera.

Florida Atlantic University, Davie | Student Government-House Speaker | Nov 2023 - Sept 2024

- Preside over House meetings, ensuring adherence to FAU Student Government Constitution, Statutes, and Robert's Rules of Order while facilitating communication for bill sponsorships.
- Chair the allocation of a \$196,910 campus-wide budget, guiding funding decisions and representing student interests in administrative meetings.

Florida Atlantic University, Dania Beach | Lab Assistant | May 2024 - July 2024

- Integrated ZED2 stereo camera with ROS2 for real-time visualization and resolved hardware/software issues.
- Annotated and prepared custom datasets with LabelImg for a YOLOv9 model for specific object detection.

Volvo India Private Limited, India | Project Trainee | Aug 2021 - Sep 2021

- Researched emerging technologies for warehouse management and explored API security protocols.
- Presented research findings and demonstrated proficiency in technical research and analysis.

SKILLS:

- **Programming Languages:** Python, C, C++, Java, R, ROS
- **Web Technologies:** HTML, MySQL
- **Microsoft Office:** MS Word, MS PowerPoint, MS Excel, MS Outlook
- **Platform:** Visual Studio, Code Blocks, Android Studio, Weka, LabelImg
- **Tools & Frameworks:** YOLOv9, TensorFlow, PyTorch
- **Languages:** English, Hindi, Malayalam, Kannada

PROJECTS UNDERTAKEN:

Brain Tumor Detection in MRI:

- Developed an enhanced brain tumor detection system using superpixels, PCA, and Template-based K-means clustering, achieving 95.0% accuracy and 97.36% sensitivity.
- Reduced image processing complexity and achieved detection in 35–60 seconds, outperforming existing schemes in speed and precision.

LEADERSHIP & COMPETITIONS

Marine Robotics Club, Secretary | Florida Atlantic University | Aug 2024 – December 2024

- Organized meetings, maintained records, and coordinated club activities focused on marine robotics innovation.

RobotX International Competition | Team Competitor | Nov 2024

- Competed internationally in advanced robotics challenges, focusing on marine robotics design, testing, and innovation.